



Evidence of epidermal growth factor receptor expression in uveal melanoma: Inhibition of epidermal growth factor-mediated signalling by Gefitinib and Cetuximab triggered antibody-dependent cellular cytotoxicity

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Abstract Despite advances in surgery and radiotherapy of uveal melanoma (UM), many patients develop distant metastases that poorly respond to therapy. Improved therapies for the metastatic disease are therefore urgently needed. Expression of the epidermal growth factor receptor (EGFR), a target of kinase inhibitors and humanised antibodies in use for several cancers, had been reported.

Forty-eight human UMs were analysed by expression profiling. Signalling was tested in three EGFR expressing UM cell lines by Western blotting using phosphorylation specific antibodies for EGFR and the downstream mediators AKT (v-akt murine thymoma viral oncogene homolog) and extracellular signal-regulated kinase (ERK). Evidence for signalling in tumours was obtained through the application of a UM-specific EGF-signature. The EGFR specific kinase inhibitor, Gefitinib and the humanised monoclonal antibody, Cetuximab, were tested for their effect on EGFR signalling. Natural killer cell mediated antibody-dependent cellular cytotoxicity (ADCC) and tumour necrosis factor α (TNF- α) release was analysed for Cetuximab.

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